

Liam Arzola

✉ larzola@ucsd.edu  <https://www.linkedin.com/in/liam-arzola/>  <https://liam0215.github.io/>

EDUCATION

Ph.D. Computer Science

University of California, San Diego

GPA: 3.95, advised by Prof. Alex Snoeren and Prof. Amy Ousterhout.

09/2024 – present

B.A. Computer Science

Cornell University

GPA: 3.35

08/2023

PUBLICATIONS

Pesto: Cooking up high performance BFT Queries

The 31st Symposium on Operating Systems Principles (SOSP 2025)

Florian Suri-Payer (Cornell University), Neil Giridharan (UC Berkeley), Liam Arzola (UC San Diego), Shir Cohen Gahtan (Cornell University), Lorenzo Alvisi (Cornell University), Natacha Crooks (UC Berkeley).

Virtuoso: High Resource Utilization and μ s-scale Performance Isolation in a Shared Virtual Machine TCP Network Stack

Under Submission

Matheus Stolet (MPI-SWS), Liam Arzola (UC San Diego), Simon Peter (University of Washington), Antoine Kaufmann (MPI-SWS).

RESEARCH EXPERIENCE

Graduate Student Researcher at UC San Diego

With Prof. Amy Ousterhout and Prof. Alex Snoeren

Investigating efficient microsecond-scale software offload engines for on-chip accelerators.

- Helped integrate Intel's In-Memory Analytics (IAA) and QuickAssist Technology (QAT) accelerators for compression and elliptic curve point multiplication offload into a novel offload engine.
- Modified Intel's Query Processing Library (QPL) to achieve more efficient accelerator offloading and polling.
- Built microbenchmarks and end-to-end benchmarks for (de)compression and ed25519 signature verification.
- Adapted the Solana validator to use offloaded point multiplication for ed25519 signature verification.

09/2024 – Present

Research Intern at The Max Planck Institute for Software Systems

Operating Systems Group, Prof. Antoine Kaufmann

Worked with Prof. Antoine Kaufmann on host network stack research.

- Worked on extending TAS, an optimized userspace TCP stack, by adding optimizations for larger flows, including hardware offload and extended message buffer support (DPDK).
- Worked on Virtuoso, a novel host networking stack for virtual machines and containers for high performance and performance isolation. Virtuoso is currently under submission (<https://arxiv.org/abs/2309.14016>).
- Worked on modifying TAS to support microsecond-scale reconfigurable optical networks in collaboration with Prof. Yiting Xia. The goal of this project was to understand the trade-offs between the increased cost of tracking more out-of-order packets on the hosts, and the performance benefit of reducing retransmissions for varying degrees of reordering.

06/2023 – 12/2023

Research Assistant at Cornell University

Theory Meets Practice Lab, Prof. Lorenzo Alvisi

I worked on a byzantine fault tolerant relational database. I designed our relational frontend engine, implemented popular online transaction processing benchmarks (TPCC/Auctionmark), and integrated CockroachDB as a baseline system. (C++)

10/2022 – present

WORK EXPERIENCE

Software Engineer

01/2024 – 08/2024

Microsoft

Worked in Azure Edge and Platform's Core OS group. Contributed to VMBus, a high performance communication channel for host to VM communication, and openvmm/openhcl, a novel virtual machine manager and hardware compatibility layer to enable unmodified guests to use SoC-accelerated networking and storage. (<https://github.com/microsoft/openvmm> ). (C, Rust)

Software Engineer Intern

05/2022 – 08/2022

Microsoft

Worked on the Azure Storage Capacity Management team on designing and implementing a leader based crash fault tolerant consensus library (Raft) for partitioning requests over a cluster of workers. (C#, .NET)

Software Engineer Intern

12/2021 – 05/2022

Quantum Resistant Ledger

Rewrote the core library from C++ to Rust and added multithreading support for concurrent XMSS signature verification.

Software Engineer Intern

06/2021 – 08/2021

Dropbox 

Designed and implemented sync progress aggregation for backups. (Rust)

TEACHING EXPERIENCE

Undergraduate Teaching Assistant - Operating Systems Practicum

01/2023 – 05/2023

Cornell University

- Teaching assistant alongside Prof. Robbert van Renesse for the optional advanced Operating Systems Practicum.
- Held office hours and graded student OS assignments in C in the Earth and Grass Operating System (EGOS) and EGOS-2000 .

SERVICE

Volunteer at The 29th ACM Symposium on Operating Systems Principles (SOSP)

10/2023